

Samyak Jain

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Education

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science (GPA: 4.0/4.0)

Aug 2025 – May 2027

Graduate Teaching Assistant: Machine Learning (Summer 2026) | Data and Visual Analytics (Fall 2026)

Specialization: Machine Learning & Computing Systems

Vellore Institute of Technology

Vellore, India

Bachelor of Technology in Computer Science (CGPA: 9.57/10, Rank 9 of 2,200+)

Aug 2018 – Jun 2022

Experience

Georgia Institute of Technology

Jan 2026 – Present

Graduate Research Assistant (LLM Systems and RAG Infrastructure)

Atlanta, GA

- Building a multi-tenant RAG evaluation platform comparing flat-vector and agentic retrieval, with tenant-isolated indexes, benchmark tracing, and automated evaluation across PostgreSQL QA and MuSiQue.
- Evaluating bounded agentic retrieval loops and failure modes across OpenAI-compatible and local vLLM backends. Current experiments achieve 51.7% PostgreSQL QA accuracy versus 23.3% for flat-vector retrieval.

Goldman Sachs

Aug 2022 – Jul 2025

Software Engineer

Bengaluru, India

- Developed backend services, investor data models, and reporting platforms supporting post-trade data workflows and client reporting within Strategic Client Services Technology in Asset & Wealth Management.
- Led the full-stack modernization of Client Service Desktop (Vert.x/Angular → Spring Boot/React), increasing developer velocity by 60% and cutting deployment time from 2 hours to 15 minutes.
- Architected and helped implement an AWS reporting platform (Step Functions, ECS, Lambda), automating ~100 templates and reducing recurring manual steps by 70% through configurable workflows and failure monitoring.
- Modeled investor positions, transactions, and accounts from heterogeneous vendor feeds in Snowflake. Implemented and owned REST APIs exposing validated & opinionated views and handling 10M+ records monthly at 99.9% availability.
- Co-founded a cross-functional architecture risk pod reviewing ~15 microservices for scalability, resiliency, and failure handling. Identified bottlenecks and drove remediation with service owners.
- Selected for the Bar Raiser in Training program, conducting design and architecture reviews to identify scalability, reliability, and cost risks.

Walmart Global Tech India

Jan 2022 – Jul 2022

Software Engineer Intern

Bengaluru, India

- Developed and deployed React UI components for STRIDE, Walmart's internal transportation platform. Supported rollouts to Chile and Mexico, then owned the module's remaining work for the U.S. rollout.
- Implemented integration tests with 90% coverage, preventing 8 potential production failures during cross-team deployments. Received an Excellence Award and full-time return offer.

Projects

High-Throughput Report Rendering Service (Goldman Sachs) | Java, Spring Boot

Aug 2023 – Sep 2023

- Designed and built a bounded-memory Java pipeline for 5GB+ JSON inputs, parallelizing row transformation with ExecutorService and preserving deterministic output through ordered reassembly, eliminating XLSX/CSV out-of-memory failures.
- Processed 100K+ rows in under 4 minutes using Jackson streaming and Apache POI SXSSF, cutting generation time by 60%. The service was adopted by 8+ teams supporting ~20K reports annually.

Hybrid Vector Search Engine & Selectivity-Aware Planner | Python, PostgreSQL, PostGIS Sep 2025 – Nov 2025

- Built an adaptive search engine over 515K hotel reviews, routing queries between global HNSW and parallel partition-local HNSW based on structured-filter selectivity.
- Combined PostgreSQL partitioning, vector indexes, and geospatial filtering to reduce average latency by 22% versus global ANN while maintaining 98.5% Recall@k.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript/TypeScript

Backend: Spring Boot, Microservices, Kafka, REST APIs

Data: PostgreSQL, MySQL, Snowflake, DynamoDB, MongoDB, Elasticsearch, Redis

AI & ML Systems: RAG, vLLM, embeddings, vector search, LLM inference and evaluation

Cloud/DevOps: AWS (ECS, Lambda, Step Functions, S3), Docker, Kubernetes, Terraform, Git

Frontend: React, Angular

Core: Data Structures & Algorithms, Object-Oriented Design, System Design, Distributed Systems, Databases, Concurrency, Performance Optimization